LivCast Capture — Complete (single sheet, Rev A)

Overlap fix: 3-81mm pitch; U701/U703 spread; I/O LCD | SDI PLAYBACK OUT (right, real symbols)

No UVC. No internal battery. Antmicro SDI→CSI detail: refs/antmicro-sdi-mipi-bridge + archive_sheets/04_SDI_CSI.kicad_sch

▯▯ REGION POWER ▯▯

REGION POWER / CM5 / LCD … see aliases at x=300

J_DTAP (VIN_RAW) | J_USBC_PWR | TPS25751D sink | LM76003 | TPS62130A | OR diodes

Rails: +5V0, +3V3, +1V8, +1V2, GND | VIN_RAW | PPHV_PD (symbols TBD — power ports below)

▯▯ REGION CM5 ▯▯

U_CM5 (Compute Module 5) — block / footprint TBD

Nets: CSI0_* (4lane+ck) | CSI1_* (4lane+ck) | HDMI_TX_* | USB_DP/DN | I2C_* | SPI_* | SDI_TX_SPI_*

CSI0 ← IT6616 (HDMI IN) | CSI1 ← CrossLink (SDI IN) | HDMI_TX → SDI PLAYBACK OUT | USB ← phone gadget

▯▯ REGION HDMI IN ▯▯

J_HDMI_IN → ESD → IT6616 → CSI0_* [IT6616 symbol TBD]

Path: HDMI Type-A in → ESD protection → IT6616 HDMI→CSI → CM5 CSI0 (4-lane + clock)

▯▯ REGION SDI IN (Antmicro path — condensed) ▯▯

SDI IN: J_SDI_IN (BNC) → GS2971A → LIF-MD6000-6JMG80I (CrossLink) → CSI1

Full detail (do not paste here): refs/antmicro-sdi-mipi-bridge/ + archive_sheets/04_SDI_CSI.kicad_sch

Note: Antmicro J2/GS2988 input-loop is NOT product SDI out — see SDI PLAYBACK OUT (right).

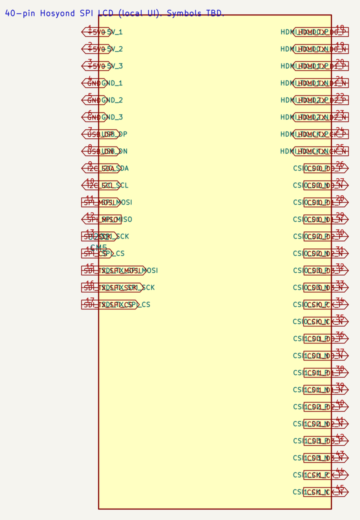
▯▯ REGION USB PHONE ▯▯

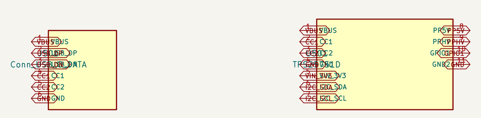
J_USBC_PHONE → TPS25751 (source/PD) → TPS25947 (eFuse) → CM5 USB gadget (DP/DN)

iPhone: USB Ethernet gadget + IP video (not UVC). Symbols TBD.

▯▯ REGION LCD ▯▯

J_LCD — 40-pin Hosyond SPI LCD (local UI). Symbols TBD.





▯▯ REGION SDI PLAYBACK OUT (from archive 07 — real symbols) ▯▯

CM5 HDMI_TX → U701 TMDS_BUF_1TO2 → J_HDMI_OUT + U702 IT66021FN → U703 GS2962A → J_SDI_OUT BNC

NET ALIAS: HDMI_TX_* globals; VIDEO[19:0] DE VS HS PCLK locals; HDMI_BUF_B_* HDMI_MON_*

PRODUCT SDI OUT = playback via GS2962A

Path: CM5 HDMI_TX_* --> U701 TMDS 1:2 --> J_HDMI_OUT (monitor) + U702 IT66021FN --> VIDEO[19:0] --> U703 GS2962A --> J_SDI_OUT

GS2962A integrated cable driver — GS2989 NOT required on this sheet

Decoupling: place per IT66021FN / GS2962A datasheets (not drawn — avoid Device lib dependency)

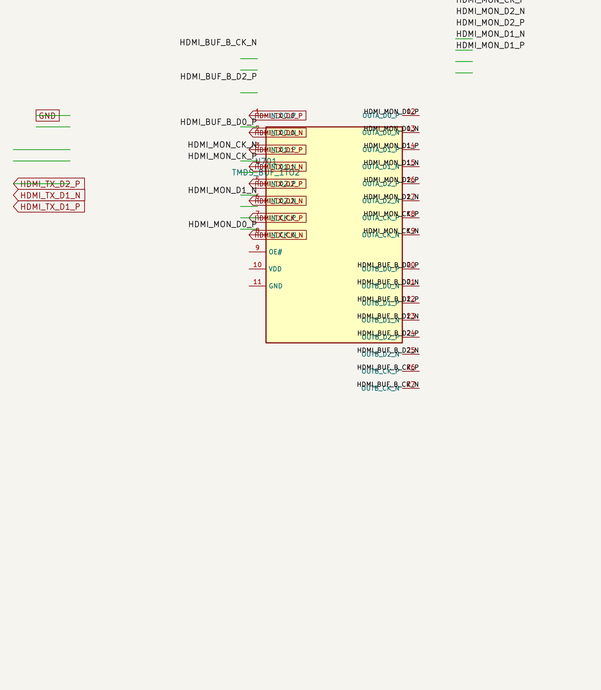
SDI out network: SDO --> AC couple + 75R to BNC center; RSET per GS2962 DS; SDO# terminate per DS; shield=GND

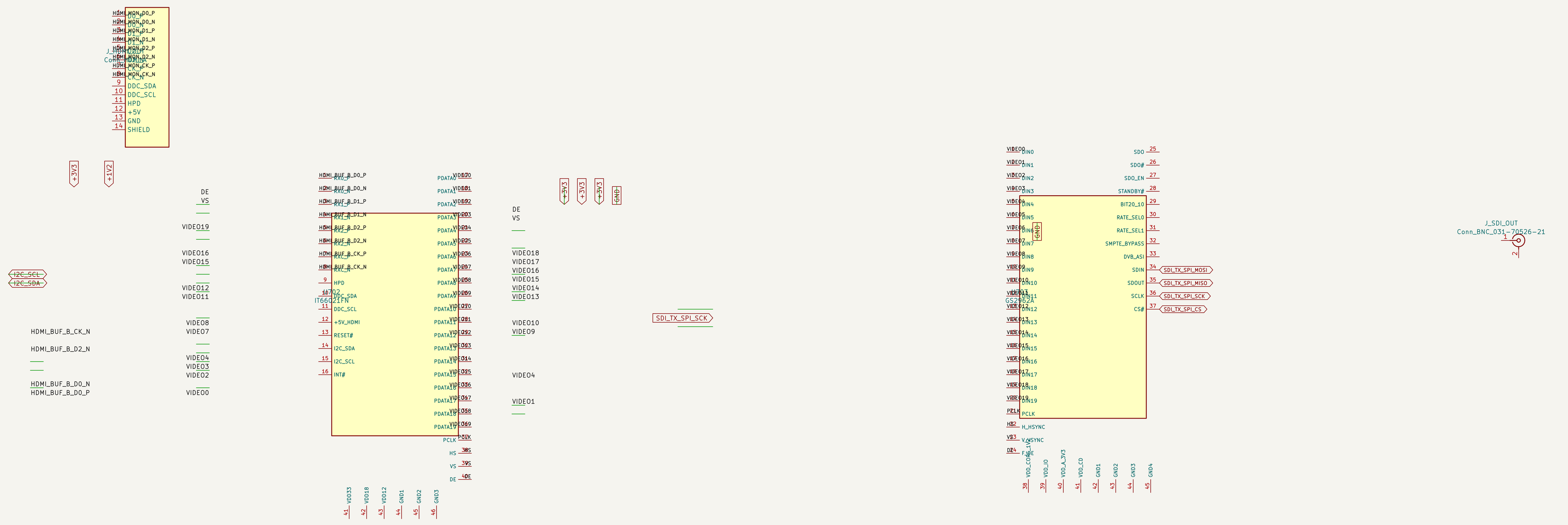
IT66021FN pinout is interface-level (pending full DS/NDA package mapping)

GSPI: SDI_TX_SPI_* from CM5 (dedicated CS); I2C_SDA/SCL to IT66021FN

Reject: Antmicro J2/GS2988 as product SDI out (see sheet 04 debug only)

Refs U701/U702/U703 avoid clash with sheet 04 Antmicro U1/U2/U3 (functional: buffer / IT66021FN / GS2962A)





SDI playback: CM5 HDMI TX → TMDS 1:2 → HDMI mon + IT66021FN → GS2962A → BNC
Single sheet. Antmicro SDI ingest detail in refs/ + archive_sheets/04

LivCast

Sheet: /
File: LivCast_Capture.kicad_sch

Title: LivCast Capture — Complete

Size: A0	Date: 2026-07-25	**Rev: A**
KiCad E.D.A. 10.0.5-rc1		Id: 1/1